

WEEKLY PROGRESS REPORT

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Report No.: Week 1

Project Title	Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors
Team Name / ID	Gujju Gang/Group 7
Team Members	Prakshal shah (AU2440265) Belisha desai (AU2440034) Kahaan shah (AU2440047) Namyaa parmar(AU2440025)
Faculty Mentor	[Mentor Name]
Repository Link	https://github.com/prakshal173/DIP_2026_17_7
Dataset	AITEX Fabric Image Database (AFID)

1. Work Done This Week

- Finalised the problem statement: adaptive fabric surface defect detection and localization using classical texture descriptors, without any deep-learning component.
- Downloaded and explored the AITEX Fabric Image Database (AFID) — verified the 245 images across 7 fabric types (140 defect-free, 105 defective across 12 defect types) and studied the accompanying ground-truth defect masks.
- Noted the unusual image geometry (4096×256 pixels, long narrow fabric strips) and discussed how this will affect the choice of sliding-window / patch size for texture analysis.
- Set up the working environment (Python, OpenCV, scikit-image) and did a quick visual inspection of a few defect-free vs. defective samples to get a feel for how subtle some defects are.
- Carried out an initial literature review to identify which classical texture-analysis techniques are commonly used for this problem (summarised in Section 3).

2. Work Planned for Next Week

- Implement a baseline preprocessing pipeline: fabric-region cropping, illumination normalisation, and splitting long strip images into fixed-size patches.
- Extract a first set of texture features (starting with GLCM statistics) on defect-free patches to establish a 'normal texture' reference.
- Run the same feature extraction on a few known defective patches and check whether simple distance-based thresholding already separates them from normal patches.
- Begin comparing single-scale vs. multi-scale window sizes, since the fabric texture period looked different across the 7 fabric types.

3. Literature Review

As this is Report 1, we reviewed prior work on classical (non-deep-learning) fabric defect detection to decide which family of texture descriptors to build on.

- **Kumar (2008), IEEE Trans. Industrial Electronics:** This survey organises fabric-defect methods into statistical, spectral (transform-based), model-based and learning-based categories. It helped us see that our project falls mainly in the statistical/spectral classical family, and that texture-deviation-from-normal is the standard way classical methods frame the problem.
- **Kumar & Pang (2002), IEEE Trans. Industry Applications:** Proposes detecting defects in textured materials using a bank of Gabor filters tuned to the fabric's dominant spatial frequency and orientation, then flagging regions whose filter response deviates from the defect-free norm. This is a strong candidate technique for the 'adaptive' part of our project, since filter parameters can be tuned per fabric type.
- **Bodnarova, Bennamoun & Latham (2002), Pattern Recognition:** Extends the Gabor-filter idea by proposing a method to automatically select the optimal filter frequency and orientation for a given fabric texture instead of fixing it manually, which is directly relevant since AFID has 7 different fabric types with different weave patterns.
- **Ngan, Pang & Yung (2011), Image and Vision Computing:** A broader review of automated fabric-defect detection that compares structural, statistical, spectral, model-based and hybrid approaches, and discusses their trade-offs on patterned vs. plain fabrics. This is guiding our decision to combine a statistical descriptor (GLCM/LBP) with a multi-scale strategy rather than relying on one fixed technique.
- **Silvestre-Blanes et al. (2019), AUTEX Research Journal:** This is the paper that introduces the AITEX/AFID dataset itself, describing how the images were captured, the 12 defect categories, and baseline results with a few classical methods — useful both as our dataset source and as a reference point for comparing our own results.

4. Challenges Faced

- The 4096×256 image size is unusual to work with directly; we are still deciding the best patch/window size so that defects are neither missed (window too large) nor mistaken for texture noise (window too small).

- Some defect types in AFID are visually very subtle even to us, which suggests a single fixed threshold will not generalise — reinforcing the need for the adaptive, per-fabric-type approach planned for next week.
 - None of us had prior hands-on experience with Gabor filters or GLCM, so part of this week went into understanding the underlying math before deciding which descriptor to implement first.
-